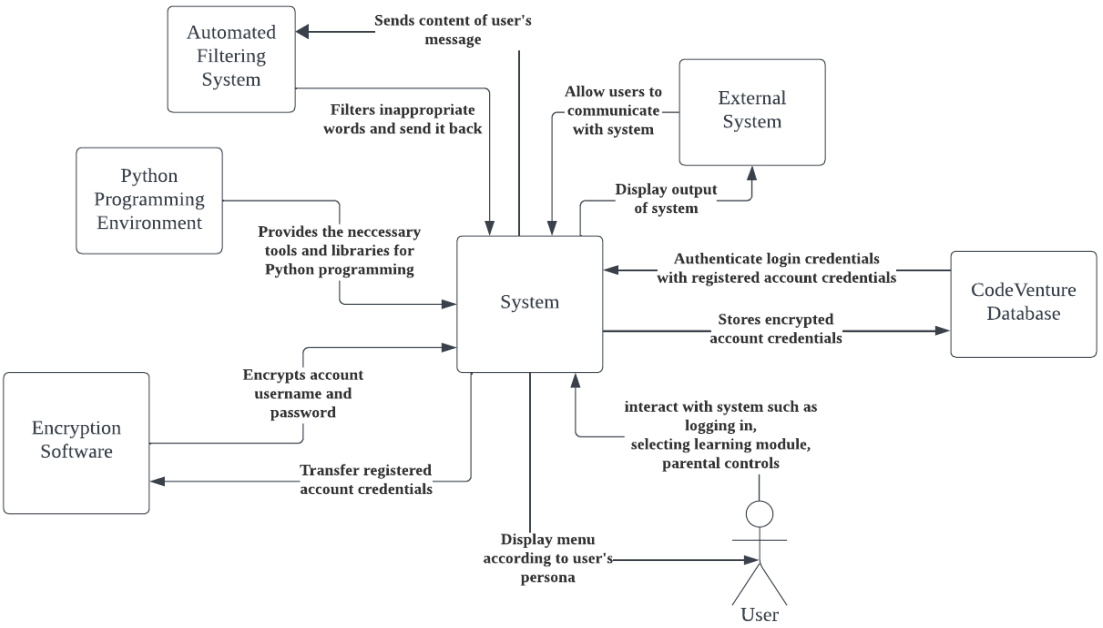
SOFTWARE DESIGN

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Comprehensive System Context Analysis

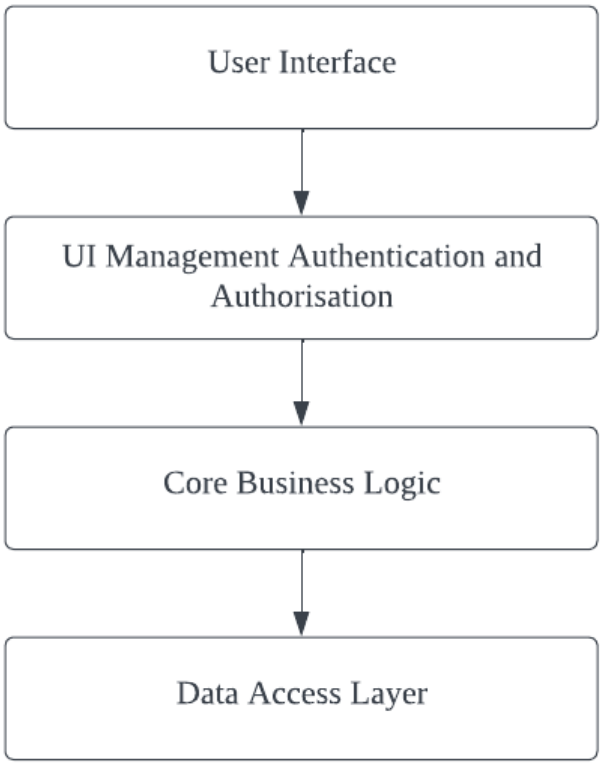


Context	Analysis
Operational	- CodeVenture Database may get hacked, causing users' login credentials to get stolen. By storing encrypted credentials and authenticating users appropriately, the chances of unauthorised access will lower
Behavioral	- One important task is to consistently update the Python Programming Environment, which will provide state of the art tools and libraries for the learners.
Ecological	- Developers are expected to constantly maintain and update CodeVenture. One way to do so is to consistently create new learning

	modules for the learners.
Sociocultural	- Users expect their accounts to be safe. Lack of safety will deter users from using CodeVenture. Encryption Software keeps accounts safe by encrypting the users' credentials. - Parents expect CodeVentue to be a safe environment for their children. Allowing inappropriate messages to show up frequently will deter parents from allowing their children to use CodeVenture. Automated Filtering System allows inappropriate messages to be censored and keeps CodeVenture safe for younger learners.

System Architecture Diagram

SAD - Layered Diagram



User interface -
Provide users a way to interact with the system such as logging in (functional requirement #1 to #6) or navigating through the system menu

UI Management Authentication And Authorisation -

To satisfy function requirement #6, which authenticates the user's login details by comparing them to existing account details stored in the database (authorises user if successfully authenticated)

Core Business Logic -
After successfully logging in, display a menu which varies according to users.

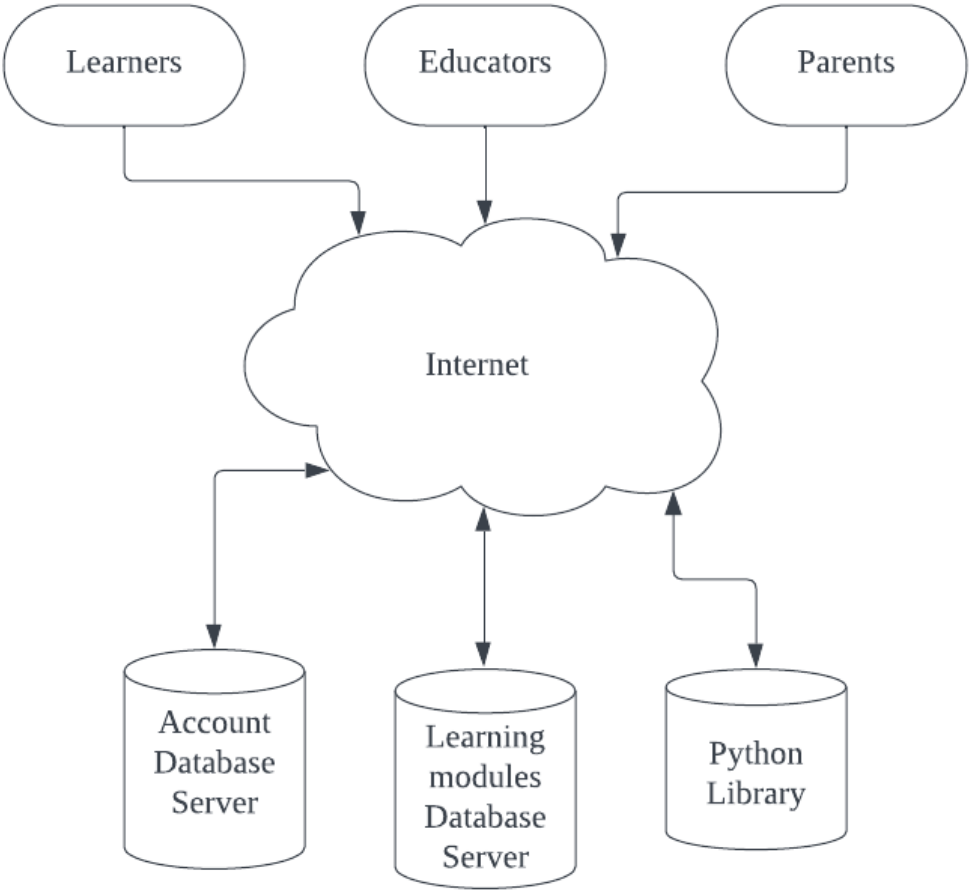
Learners may pick learning modules that intrigue them - (functional requirement #13 and #14)

Educators can track and assist with learners' progress

Parents can moderate and limit learners' gameplay hours if required

Data Access Layer - The layer that stores registered user credentials and learning modules.

SAD - Client - Server Architecture



Rationale for client - server architecture -

The server handles data management leaving the client more processing power to work on other tasks. This satisfies NFR #1 to #3.

Rationale for storing account details and learning modules in different servers -

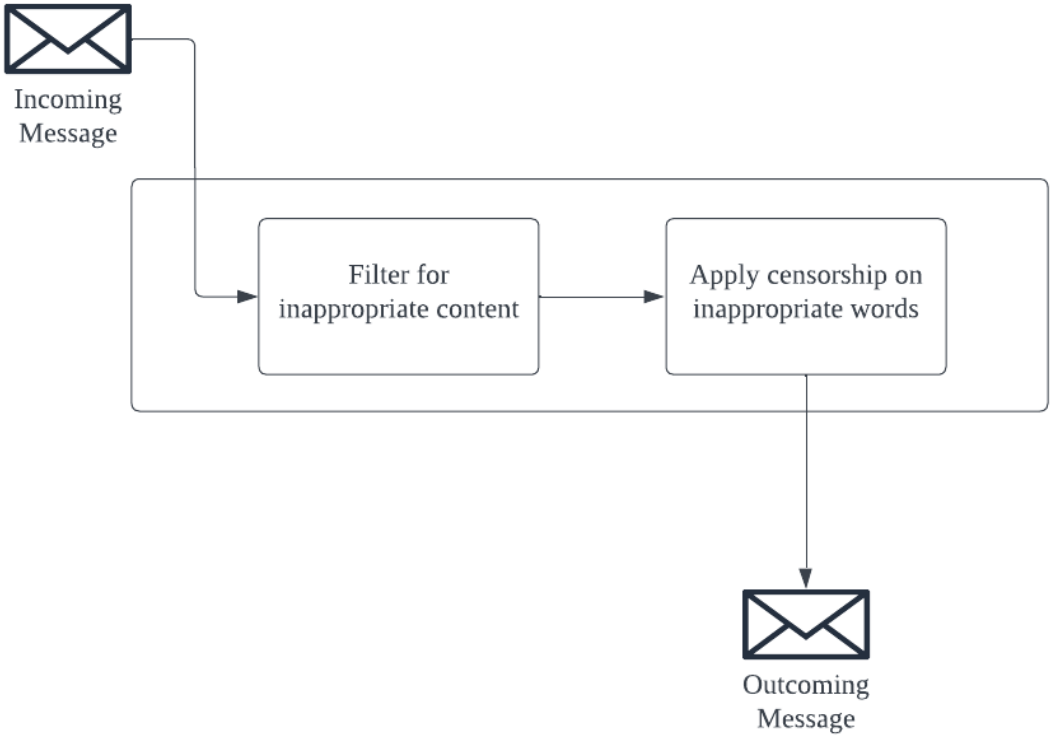
As learning modules are accessible by users and can be displayed by the system, it is not desirable to store account details in the same database server as learning modules.

Rationale for Python library server -

CodeVenture’s purpose is to allow users to learn Python programming language, hence, access to Python library server is required to install the necessary interpreter and tools.

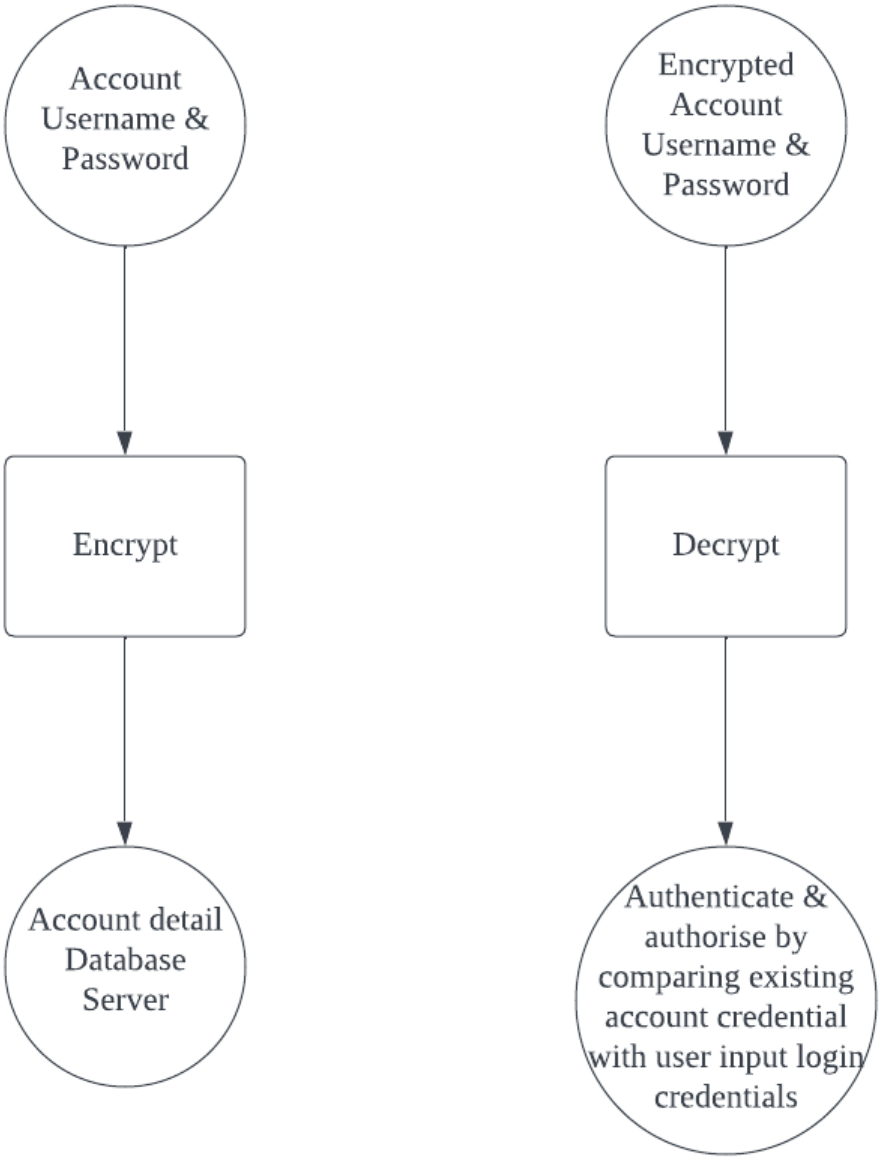
SAD - Pipe & Filter Architecture Pattern

Automated Filter Tool



When a message is sent, its content is scanned and inappropriate content is filtered before sending it to the recipient. This satisfies NFR #5 and #6 as it avoids exposing inappropriate content to children.

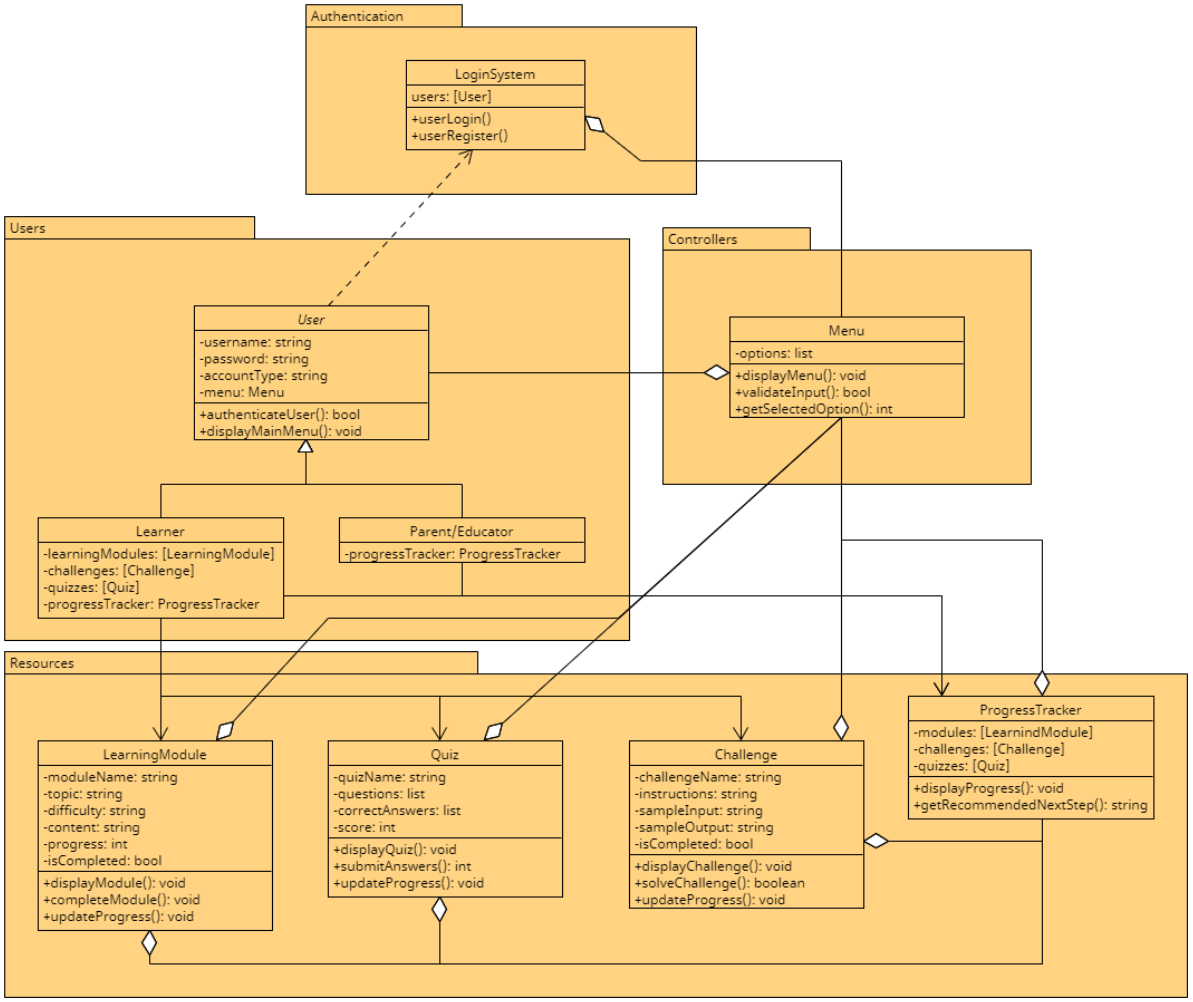
Account Security via Encryption



When a user registers an account within the system, the account credentials are encrypted and stored in the registered accounts database.

Similarly, when a user attempts to login, accounts in the database are taken out and decrypted to check if the login credential exists.
This is done so to strengthen account security.

Class Diagram

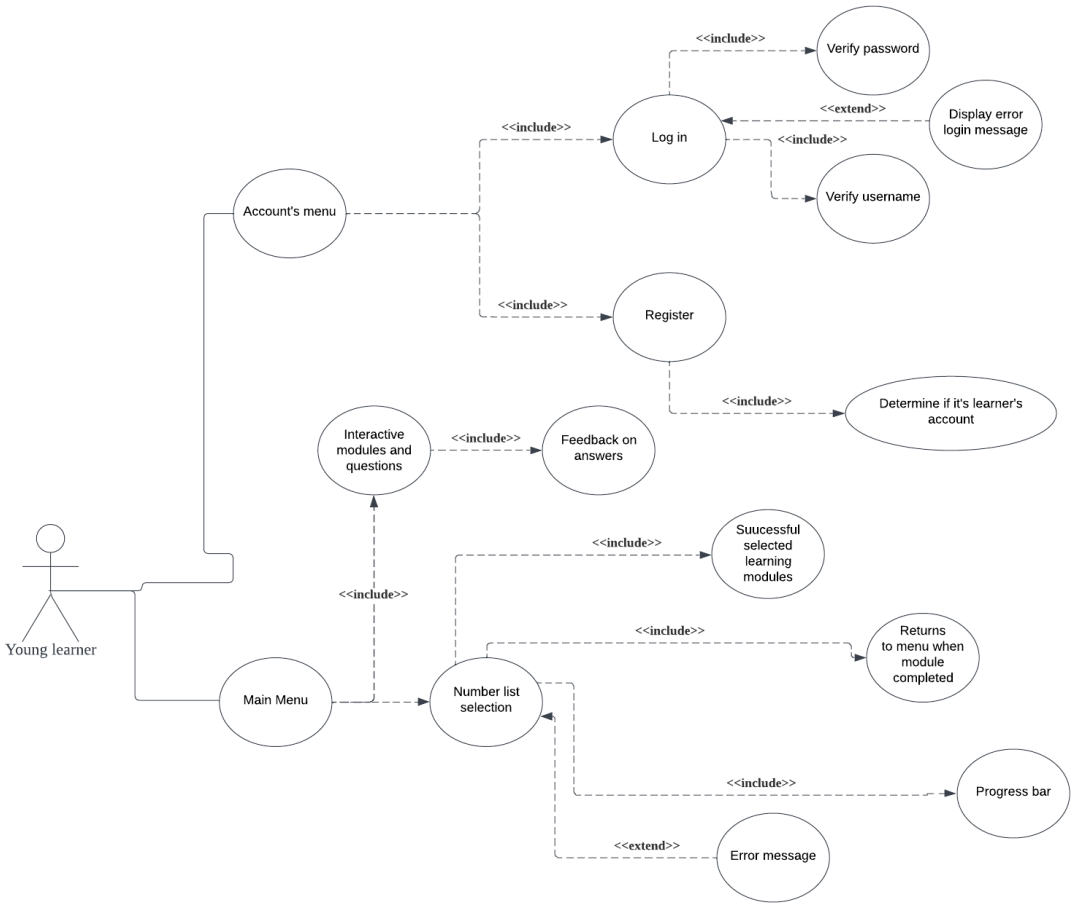


As has been shown earlier in the document, the architecture is a four-layered architecture of the following form: UI Layer -> UI Management Authentication And Authorisation -> Core Business Logic -> Data Access Layer.

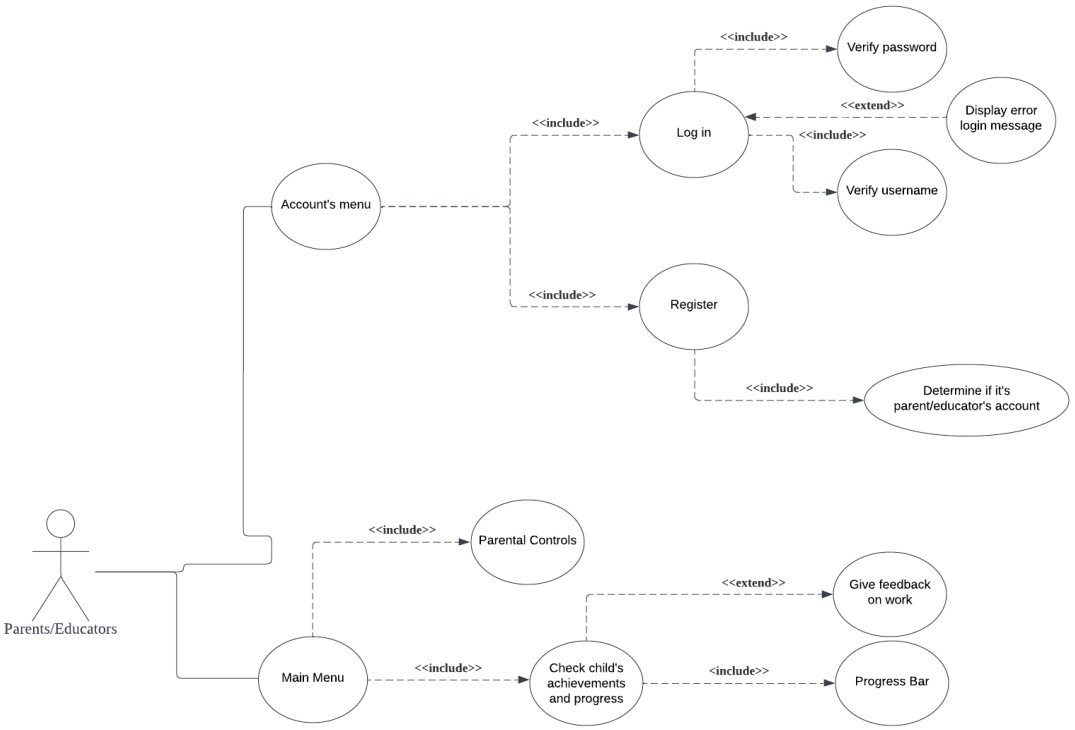
In line with the aforementioned architecture, we have implemented the Menu class to serve as the UI layer. We have also designed our system to implement a LoginSystem class which serves as the UI Management Authentication And Authorisation layer. The core business logic occurs under all the classes inside the Resources package, namely LearningModule, Quiz, Challenge and ProgressTracker.

Use Case Diagram

Young Learner Use Case Diagram

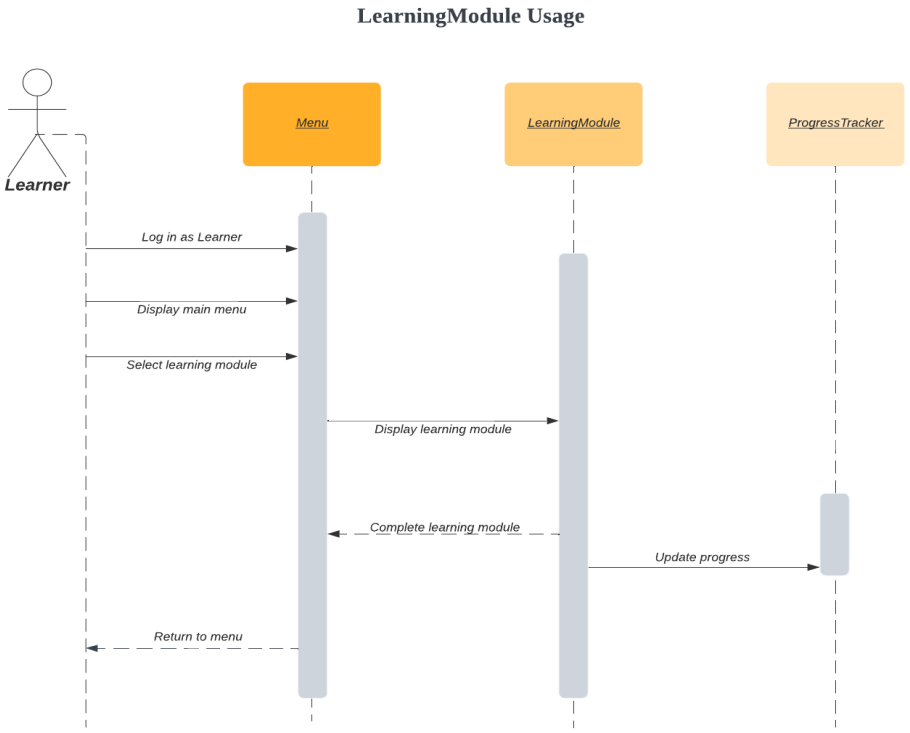


Parent / Educator Use Case Diagram

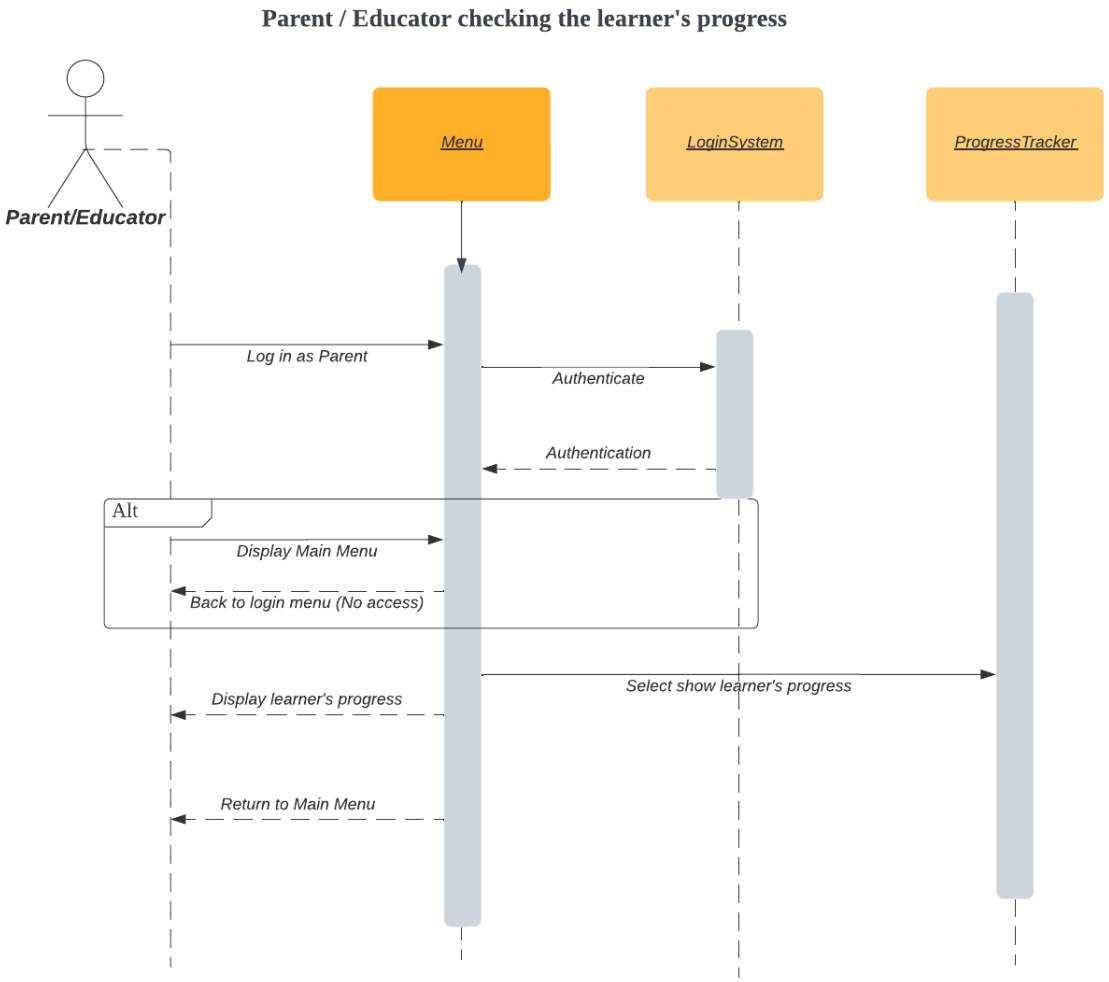


Sequence Diagram

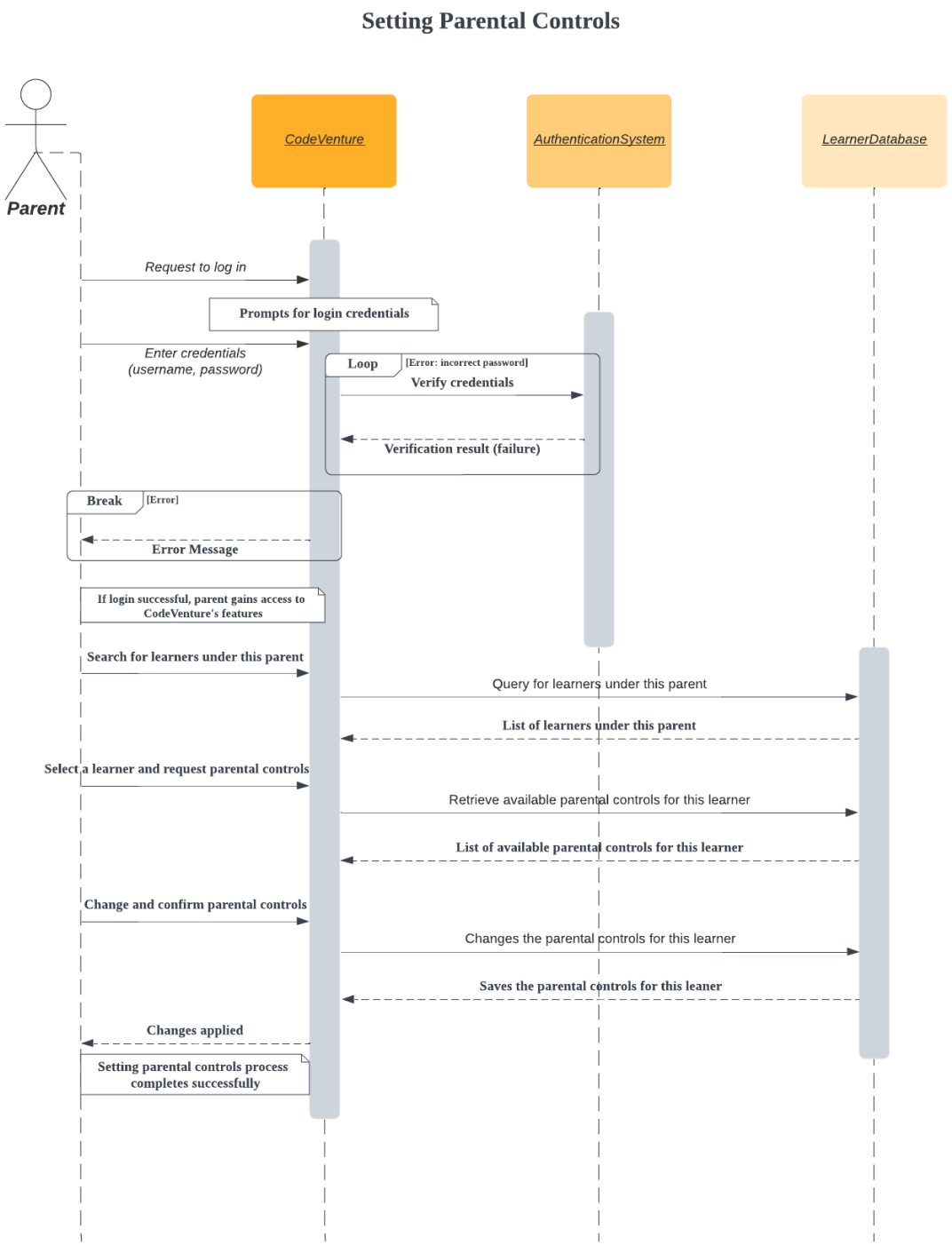
Usage of Learning Modules



Parent / Educator Stakeholder View



Setting Parental Controls



Educator Providing Feedback

